# MA321 Review 3 Answers

**1.** % Error = Abs(Difference) ÷ Actual.

| Month | Actual | Forecast | Difference | Abs(Difference) | % Error |
|---|---|---|---|---|---|
| 1 | 50 | 45 | 5 | 5 | 10% |
| 2 | 80 | 88 | −8 | 8 | 10% |
| 3 | 40 | 50 | −10 | 10 | 25% |

MAPE = (10% + 10% + 25%) ÷ 3 = **15%**

**2.**

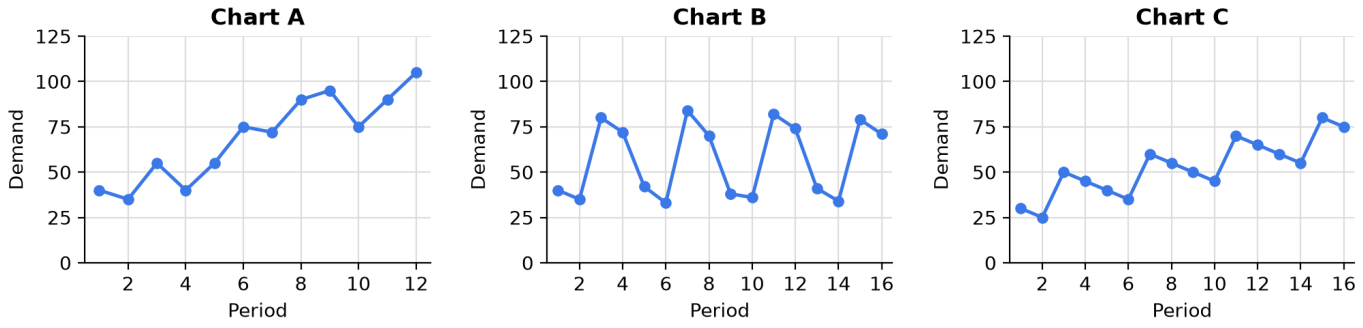



Chart A: upward trend, not seasonal, lots of noise.
Chart B: no trend (stationary), seasonal (repeats every 4 periods), little noise.
Chart C: upward trend, seasonal (repeats every 4 periods), little noise.

a) **True.** Chart B has no trend; it moves up and down around the same level.

b) **False.** Chart A has no repeating pattern; its ups and downs are noise.

c) **True.** Chart C goes up over time.

d) **True.** Chart A jumps around a lot; Chart C follows a clean repeating pattern.

e) **False.** Chart C has an upward trend, so it is not stationary.

**3.** Forecast = baseline × seasonal index.

a) Baseline 20: 20 × 0.6 = **12**, 20 × 1.2 = **24**, 20 × 0.95 = **19**, 20 × 1.1 = **22**, 20 × 1.4 = **28**, 20 × 0.75 = **15**

b) 20% lower baseline = 20 × 0.8 = 16: **9.6, 19.2, 15.2, 17.6, 22.4, 12** (each forecast in a is 20% lower too)

**4.**

a) **Forecast 2.** The lower MAPE (12%) is more accurate.

b) The forecast graph stays very close to the actuals graph. On average the forecast is only 3% off from the actual demand.

c) **True.** If the forecasts are far off (for example, forecasting 30 when the actual is 10 is a 200% error), the MAPE can be over 100%.

**5.**

a) Average of the composites = (30 + 45 + 60 + 65) ÷ 4 = 200 ÷ 4 = 50. Index = composite ÷ 50.

Quarter 1: $30 \div 50 = \mathbf{0.6}$
Quarter 2: $45 \div 50 = \mathbf{0.9}$
Quarter 3: $60 \div 50 = \mathbf{1.2}$
Quarter 4: $65 \div 50 = \mathbf{1.3}$

b) Forecast = $40 \times$ index: Q1 = **24**, Q2 = **36**, Q3 = **48**, Q4 = **52**